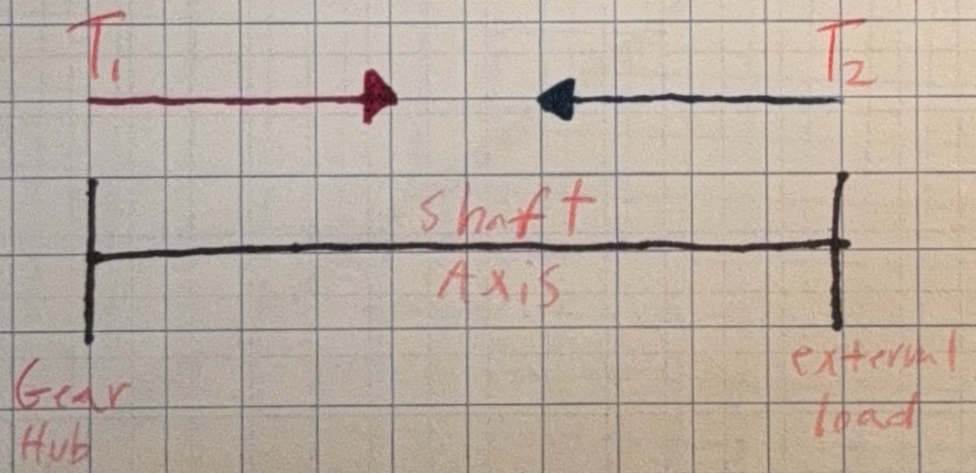
Output Shaft

$$\tau_{max} = \frac{Tr}{J}, \quad r = \frac{d}{2}, \quad J = \frac{\pi d^4}{32}$$

$$\tau_{max} = \frac{16\,T}{\pi d^3} = \frac{16(1.25)}{\pi (0.250)^3}$$

$$\boxed{\tau_{max} = 407.44 \text{ psi}}$$

FBD:



$$T_1 = T_2 = 1.25 \text{ in} \cdot \text{lbf}$$

$$\Sigma T = 0$$

407.44 psi $<$ 8,500 psi